The Toroidal Grid $C_3 \,\square\, C_n$ Is 3-Choosable for Every n

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Abstract

Cai, Wang and Zhu conjectured that every toroidal grid $C_m \,\square\, C_n$ is 3-choosable. Li, Shao, Petrov and Gordeev proved this when m or n is even, and showed that the Alon–Tarsi number of $C_m \,\square\, C_n$ is 4 when m and n are both odd, so that the Alon–Tarsi method cannot reach the remaining case. We prove that $C_3 \,\square\, C_n$ is 3-choosable for every odd n. Hence $\chi_\ell(C_3 \,\square\, C_n) = 3$ for every $n \geq 3$. The proof cuts the torus open along two columns, colours the remaining strip with the Combinatorial Nullstellensatz while pinning one colour and forbidding one other, and then fills the two columns by hand; the forbidden colour is chosen in advance so that the fill cannot fail. We also record a second, kernel-method proof for the family $n = 3k$, k odd, and what is known so far for $C_m \,\square\, C_n$ with $m, n \geq 5$ odd.

Status of this draft. Every step of the proof of Theorem A is either proved in full here or is a finite statement checked by exhaustive computation over all list assignments up to renaming of colours (Lemma 4.1). The scripts are listed in Section 8. Still to do before submission: (i) a hand proof of Lemma 4.1 (Amrutanshu's prism lemma has one, by Hall's theorem, to be merged); (ii) a full literature check of the papers citing [LSPG] and of the dynamic survey [S], to confirm that the odd case of $C_3 \,\square\, C_n$ is not already settled.

1 Introduction

All graphs are finite and simple. A *list assignment* L of a graph G gives each vertex v a set $L(v)$ of colours. An *L-colouring* is a proper colouring c with $c(v) \in L(v)$ for every v. The graph G is *k-choosable* if it has an L-colouring whenever $|L(v)| \geq k$ for all v, and the *choice number* $\chi_\ell(G)$ is the least such k.

For $m, n \geq 3$ the *toroidal grid* $C_m \,\square\, C_n$ has vertex set $\mathbb{Z}_m \times \mathbb{Z}_n$, and (i, j) is adjacent to $(i \pm 1, j)$ and to $(i, j \pm 1)$. We call i the *row* and j the *column*. Column j induces a cycle of length m; row i induces a cycle of length n. When $m = 3$ every column is a triangle, so $\chi_\ell(C_3 \,\square\, C_n) \geq 3$.

Cited Theorem 1.1 (Li, Shao, Petrov and Gordeev [LSPG])**.** *If m or n is even then $C_m \,\square\, C_n$ is 3-choosable. If m and n are both odd then the Alon–Tarsi number of $C_m \,\square\, C_n$ equals* 4.

So for odd m, n the conjecture of Cai, Wang and Zhu [CWZ] is open, and the polynomial method on its own cannot settle it. The smallest case $C_3 \,\square\, C_3 = K_3 \,\square\, K_3$ is the line graph of $K_{3,3}$, and Galvin's theorem [G] shows that it is 3-choosable.

Theorem A. *For every odd $n \geq 3$ the toroidal grid $C_3 \,\square\, C_n$ is 3-choosable. Consequently $\chi_\ell(C_3 \,\square\, C_n) = 3$ for every $n \geq 3$.*

The second sentence follows from the first, from Cited Theorem 1.1 for even n, and from the triangles. The case $n = 3$ is Galvin's theorem. For odd $n \geq 5$ the proof is in Section 5. It uses two tools: an *odd strip lemma* (Section 3), proved with the Combinatorial Nullstellensatz, and a *two-column lemma* for the triangular prism (Section 4).

2 Tools from the polynomial method

For a graph G with an orientation of its edges, the *graph polynomial* is $P_G = \prod_{(u,v)}(x_u - x_v)$, the product over all arcs.

Cited Theorem 2.1 (Combinatorial Nullstellensatz, Alon [A]). *Let $P \in \mathbb{F}[x_1, \dots, x_N]$ have degree $\sum_i t_i$, and suppose the coefficient of $\prod_i x_i^{t_i}$ in P is nonzero. If $S_1, \dots, S_N \subseteq \mathbb{F}$ with $|S_i| > t_i$, then some $s \in S_1 \times \cdots \times S_N$ has $P(s) \neq 0$.*

Applied to P_G with $|L(v)| \geq t_v + 1$ (colours viewed as complex numbers), a point where $P_G \neq 0$ is an L-colouring.

Cited Theorem 2.2 (Coefficient formula, Lasoń [La], Karasev and Petrov [KP]). *Let P have degree $\sum_i t_i$ and let $A_i \subseteq \mathbb{F}$ with $|A_i| = t_i + 1$. Put $\varphi_i(z) = \prod_{b \in A_i}(z - b)$. Then*

$$[x_1^{t_1} \cdots x_N^{t_N}]\, P \;=\; \sum_{a \in A_1 \times \cdots \times A_N} \frac{P(a)}{\varphi_1'(a_1) \cdots \varphi_N'(a_N)}.$$

Throughout, $\omega = e^{2\pi i/3}$ and

$$g \;=\; 3(\omega - \omega^2) \;=\; 3i\sqrt{3}, \qquad g^2 = -27.$$

A direct expansion gives $(1-\omega)^3 = -3\omega + 3\omega^2 = -g$ and $(1-\omega^2)^3 = 3\omega - 3\omega^2 = g$. Define $\chi : \mathbb{Z}_3 \to \{0, \pm 1\}$ by $\chi(0) = 0$, $\chi(1) = -1$, $\chi(2) = 1$. Then

$$(1 - \omega^d)^3 = g\,\chi(d) \qquad (d \in \mathbb{Z}_3). \tag{1}$$

3 The odd strip lemma

For $\ell \geq 2$ let $H_\ell = C_3 \,\square\, P_\ell$, with vertex set $\mathbb{Z}_3 \times \{1, \dots, \ell\}$: each column is a triangle and row i is the path $(i,1), \dots, (i,\ell)$. It has 3ℓ vertices, 3ℓ column edges and $3(\ell-1)$ row edges, so $6\ell - 3$ edges.

An *end weight* is a map $\mu : V(H_\ell) \to \{0,1,2\}$, zero outside columns 1 and ℓ, with $\sum_v \mu(v) = 3$. Put

$$k = \sum_i \mu(i,\ell), \qquad s = \sum_v \mu(v)\,\mathrm{row}(v) \bmod 3.$$

Lemma 3.1 (Odd strip lemma). *Let $\ell \geq 3$ be odd and let μ be an end weight with $k \in \{1,2\}$ and $s \not\equiv 0 \pmod 3$. Then H_ℓ has an L-colouring for every list assignment with $|L(v)| \geq 3 - \mu(v)$ for all v.*

So one may *pin* a vertex of weight 2 (list of size 1) and *forbid* one colour at a vertex of weight 1 (list of size 2).

Proof. Orient each column edge $(i,c) \to (i+1,c)$ and each row edge $(i,c) \to (i,c+1)$, and let P be the graph polynomial. Put $t_v = 2 - \mu(v) \geq 0$. Then $\sum_v t_v = 6\ell - 3 = \deg P$, so by Cited Theorem 2.1 it suffices to show that $c_\mu := [\prod_v x_v^{t_v}]\, P \neq 0$.

(1) *Reduction to cube roots.* Let $Q = P \cdot \prod_v x_v^{\mu(v)}$. Then $\deg Q = 6\ell$ and $c_\mu = [\prod_v x_v^2]\, Q$. Apply Cited Theorem 2.2 to Q with every $t_v = 2$ and every $A_v = \{1, \omega, \omega^2\}$. Here $\varphi(z) = z^3 - 1$ and $\varphi'(a) = 3a^2 = 3a^{-1}$ for $a^3 = 1$. Hence

$$c_\mu = 3^{-3\ell} \sum_{\phi} P(\phi) \prod_v \phi(v)^{1+\mu(v)},$$

summed over all maps $\phi : V(H_\ell) \to \{1, \omega, \omega^2\}$.

(2) *Column states.* $P(\phi) = 0$ unless ϕ is proper, and in particular unless ϕ is injective on every column. The bijections $\mathbb{Z}_3 \to \mathbb{Z}_3$ are exactly the six maps $i \mapsto a + \varepsilon i$ with $a \in \mathbb{Z}_3$, $\varepsilon = \pm 1$. So the contributing ϕ have the form $\phi(i, c) = \omega^{a_c + \varepsilon_c i}$.

(3) *Column edges.* For one column, $\prod_{i \in \mathbb{Z}_3} (\omega^{a+\varepsilon i} - \omega^{a+\varepsilon(i+1)}) = \omega^{3a+3\varepsilon}(1 - \omega^{\varepsilon})^3 = (1 - \omega^{\varepsilon})^3 = -\varepsilon g$.

(4) *Row edges.* Take columns $c, c+1$ with states (a, ε) and (b, ε'). If $\varepsilon' = -\varepsilon$, the colours of (i, c) and $(i, c+1)$ agree when $a + \varepsilon i \equiv b - \varepsilon i$, that is, for $i \equiv 2\varepsilon(b - a)$. So some row edge has equal ends and the product is 0. If $\varepsilon' = \varepsilon$, the product over the three row edges is $\prod_i \omega^{a+\varepsilon i}(1 - \omega^{b-a}) = (1 - \omega^{b-a})^3 = g\,\chi(b - a)$ by (1). Hence $\varepsilon_c = \varepsilon$ is constant, and with $d_c = a_{c+1} - a_c$,

$$P(\phi) = (-\varepsilon g)^{\ell}\, g^{\ell - 1} \prod_{c=1}^{\ell - 1} \chi(d_c).$$

(5) *Vertex weights.* In column c, $\sum_i (a_c + \varepsilon i) = 3a_c + 3\varepsilon \equiv 0$, and $\sum_i (a_c + \varepsilon i)\mu(i, c) = a_c M_c + \varepsilon R_c$ with $M_c = \sum_i \mu(i, c)$ and $R_c = \sum_i i\,\mu(i, c)$. Now $M_1 = 3 - k$, $M_\ell = k$, the other M_c vanish, and $R_1 + R_\ell \equiv s$. Since $3a_1 \equiv 0$,

$$\prod_v \phi(v)^{1+\mu(v)} = \omega^{k(a_\ell - a_1) + \varepsilon s}.$$

(6) *Summation.* The map $(a_1, \ldots, a_\ell) \mapsto (a_1, d_1, \ldots, d_{\ell-1})$ is a bijection and $a_\ell - a_1 = \sum_c d_c$. So

$$c_\mu = 3^{-3\ell}\, g^{2\ell - 1} \Big(\sum_{\varepsilon = \pm 1} (-\varepsilon)^{\ell} \omega^{\varepsilon s} \Big) \cdot 3 \Big(\sum_{d \in \mathbb{Z}_3} \chi(d)\, \omega^{kd} \Big)^{\ell - 1}.$$

For odd ℓ the first bracket is $\omega^{-s} - \omega^{s} = \omega^{2s} - \omega^{s}$, which is nonzero because $s \not\equiv 0$. The second bracket is $\mu_k := \omega^{2k} - \omega^{k}$, which is nonzero because $k \not\equiv 0 \pmod 3$. Since $g \neq 0$, $c_\mu \neq 0$. □

Remark 3.2. Simplifying with $g^2 = -27$ gives $c_\mu = \frac{g}{9}\,\mu_k^{\ell - 1}(\omega^{2s} - \omega^{s}) = \pm 3^{(\ell-1)/2}$ for odd ℓ, $k \in \{1, 2\}$, $s \not\equiv 0$. This closed form (for every $\ell \geq 2$ and every end weight) was checked against three independent exact computations for ℓ from 2 to 11, with no mismatch (script `ws-W1`). The lemma and the computation first appear in Amrutanshu's notes; the proof above is the one we re-derived.

4 The two-column lemma

The *prism* $\Pi = K_3 \,\square\, K_2$ has two triangles $u_0u_1u_2$ and $v_0v_1v_2$ and the three edges u_tv_t.

Lemma 4.1 (Two-column lemma)**.** *Let $A_0, A_1, A_2, B_0, B_1, B_2$ be lists for $u_0, u_1, u_2, v_0, v_1, v_2$ with $|A_0| = 3$ and every other list of size 2 or 3. If Π has no colouring from these lists, then $|B_1| = |B_2| = 2$ and $B_1 = B_2$.*

Proof (exhaustive computation). Colourability does not change when colours are renamed. Read the lists in the order $A_0, A_1, A_2, B_0, B_1, B_2$ and rename colours in order of first appearance. Afterwards $A_0 = \{0, 1, 2\}$, and each later list consists of some colours already used together with an initial segment $u, u+1, \ldots$ of the unused ones, where u is the number of colours used so far. The program `prism_sizes.c` enumerates all 236 160 170 assignments of this normal form, decides colourability of each by complete search, and finds 8 335 that are not colourable. All of them have $|B_1| = |B_2| = 2$ and $B_1 = B_2$. □

Remark 4.2 (Amrutanshu's prism lemma). For sizes exactly $(3,2,2,2,2,2)$ the non-colourable assignments are exactly those of the two forms (Z) $B_0 = B_1 = B_2$, and (W) $A_1 = A_2 = D$, $A_0 = D \cup \{\zeta\}$, $B_1 = B_2 = E$, $B_0 \setminus E = \{\zeta\}$. In (W), u_1, u_2 use up D and force $u_0 = \zeta$, while v_1, v_2 use up E and force $v_0 = \zeta$. Amrutanshu proves this by hand with Hall's theorem; `prism.c` confirms it on all 397 546 normal-form assignments (986 not colourable, no exception). Both (Z) and (W) have $B_1 = B_2$, which is all the proof uses.

5 Proof of Theorem A

Let $n \geq 5$ be odd and let L be a list assignment of $T = C_3 \,\square\, C_n$ with $|L(v)| = 3$ for every v (larger lists can be shrunk). We consider two cases.

Case 1: some row list is not contained in the next one

Suppose there are $r \in \mathbb{Z}_3$, $j \in \mathbb{Z}_n$ and a colour

$$\alpha \in L(r,j) \setminus L(r,j+1).$$

Write C_i for column $j+i$, and let r', r'' be the two rows other than r.

(1) *Choice of β.* If $|L(r',C_2) \cap L(r'',C_2)| = 2$, let β be the unique colour of $L(r',C_2) \setminus L(r'',C_2)$. Otherwise let β be any colour.

(2) *The strip.* Delete columns C_1 and C_2. Since $n \geq 5$, what remains is a copy of H_{n-2} with columns $C_3, C_4, \ldots, C_{n-1}, C_0$ in this order, and $n-2 \geq 3$ is odd. Let $\mu(r, C_0) = 2$, $\mu(r', C_3) = 1$, and $\mu = 0$ elsewhere. The last column is C_0, so $k = 2$, and $s \equiv 2r + r' \equiv r' - r \not\equiv 0$.

(3) *Colour the strip.* Give (r, C_0) the list $\{\alpha\}$, give (r', C_3) the list $L(r', C_3) \setminus \{\beta\}$, and keep every other list. These satisfy $|L(v)| \geq 3 - \mu(v)$, so Lemma 3.1 gives an L-colouring ψ of the strip with $\psi(r, C_0) = \alpha$ and $\psi(r', C_3) \neq \beta$.

(4) *The lists left for $C_1 \cup C_2$.* The subgraph on $C_1 \cup C_2$ is a prism with $u_t = (t, C_1)$ and $v_t = (t, C_2)$. A vertex of C_1 has exactly one neighbour outside $C_1 \cup C_2$, namely its row neighbour in C_0; a vertex of C_2 has exactly one, its row neighbour in C_3. (These are different columns since $n \geq 5$.) So ψ extends to T if and only if the prism is colourable from

$$A_t = L(t, C_1) \setminus \{\psi(t, C_0)\}, \qquad B_t = L(t, C_2) \setminus \{\psi(t, C_3)\}.$$

All these lists have size 2 or 3, and $A_r = L(r, C_1)$ has size 3 because $\psi(r, C_0) = \alpha \notin L(r, C_1)$.

(5) *The fill cannot fail.* Suppose it does. By Lemma 4.1 (with $u_0 = (r, C_1)$ and $\{u_1, u_2\} = \{(r', C_1), (r'', C_1)\}$), $B_{r'} = B_{r''} = E$ for a 2-set E. Then $y' := \psi(r', C_3) \in L(r', C_2)$ and $y'' := \psi(r'', C_3) \in L(r'', C_2)$, so $L(r', C_2) = E \cup \{y'\}$ and $L(r'', C_2) = E \cup \{y''\}$, with $y', y'' \notin E$. Also $y' \neq y''$, since (r', C_3) and (r'', C_3) are adjacent. Hence $L(r', C_2) \cap L(r'', C_2) = E$ has size 2, and y' is the unique colour of $L(r', C_2) \setminus L(r'', C_2)$. By step (1) this colour is β, so $\psi(r', C_3) = \beta$, contradicting step (3).

So ψ extends to an L-colouring of T.

Case 2: every row list is contained in the next one

Then $L(r,j) \subseteq L(r,j+1)$ for all r and j. All lists have size 3, so $L(r,j) = L(r,j+1)$, and each row has a constant list A_r. By Galvin's theorem, $C_3 \,\square\, C_3 = L(K_{3,3})$ is 3-choosable, so it has a colouring in which every vertex of row r gets a colour from A_r. Let $X, Y, Z : \mathbb{Z}_3 \to$ colours be its three columns. Each is a proper colouring of a triangle with $X(r), Y(r), Z(r) \in A_r$, and $X(r), Y(r), Z(r)$ are pairwise distinct, since row r of $C_3 \,\square\, C_3$ is a triangle. Colour column j of T by X if $j = 0$, by Y if j is odd and by Z if $j \geq 2$ is even. Consecutive columns get (X, Y), (Y, Z) or (Z, Y), and the pair $(n-1, 0)$ gets (Z, X) because $n-1$ is even. All these pairs are compatible, so the colouring is proper.

The two cases cover every list assignment, which proves Theorem A. □

Remark 5.1 (Credit and history). The route (edge deletion, the strip lemma, the prism lemma, and a two-case proof) is due to Amrutanshu. His first version split cases on $F(j) = \bigcap_r L(r,j)$ and needed a propagation lemma, $L(r,j) \setminus F(j) \subseteq L(r,j+1)$ for bad assignments, to reach Case 2. Case 1 above only needs $\alpha \in L(r,j) \setminus L(r,j+1)$, so splitting on that directly makes the propagation lemma (and with it the edge-deletion theorem) unnecessary. This simplification is in Amrutanshu's work plan of 26 October. We checked it by hand and, before Lemma 4.1 was verified exhaustively, by testing the whole extension step on every list assignment of $C_1 \cup C_2$ over palettes of 5 and 6 colours (200 000 and 6 400 000 cases, no failure; with the choice of β removed, 14 592 failures over 5 colours).

6 A second proof for $n = 3k$, k odd: the kernel method

For $n = 3k$ there is a proof of a different kind, by Amrutanshu, which we checked line by line. Colour $C_3 \,\square\, C_{3k}$ by $\varphi(i,j) = (i+j) \bmod 3$ and orient column edges towards the smaller colour and row edges towards the larger colour. Every vertex then has out-degree exactly 2. A minimal-counterexample argument shows that this orientation is *kernel-perfect* when k is odd, and the kernel lemma of Bondy, Boppana and Siegel gives 3-choosability. Oddness of k is used once, to walk around the columns in steps of 2.

Our computations (workstream W5) agree: this orientation is kernel-perfect exactly when $k \not\equiv 2 \pmod 4$. For $k \equiv 2 \pmod 4$ the orientation contains an induced directed cycle of odd length $3n/2$. For $n = 5, 7, 11, 13$ no orientation of $C_3 \,\square\, C_n$ with all out-degrees 2 is kernel-perfect, so the kernel method is limited to $3 \mid n$. The kernel lemma also holds for paintability, so this proof gives the stronger statement that $C_3 \,\square\, C_{3k}$ is 3-paintable for odd k. The proof of Theorem A does not give paintability, because β is chosen from the lists in advance.

7 Towards $C_m \,\square\, C_n$ with $m, n \geq 5$ odd

7.1 What carries over

Corollary 7.1. *Let $m, n \geq 3$ be odd.*

(a) $C_m \,\square\, C_3$ is 3-choosable.
(b) If L is a 3-list assignment of $C_m \,\square\, C_n$ that is constant along every row, or constant along every column, then $C_m \,\square\, C_n$ is L-colourable.

Proof. (a) $C_m \,\square\, C_3 \cong C_3 \,\square\, C_m$, and Theorem A applies. (b) Say $L(i,j) = A_i$. By (a), $C_m \,\square\, C_3$ has a colouring with colours from A_i in row i; its three columns X, Y, Z are proper colourings of

the m-cycle, pairwise compatible. Colour the columns of $C_m \,\square\, C_n$ by $X, Y, Z, Y, Z, \ldots, Y, Z$ as in Case 2. Columns are symmetric. □

Theorem 7.2 (Edge deletion for $m = 5, 7$; computer-assisted, W4)**.** *Let $m \in \{5, 7\}$ and $n \geq 3$ odd. For every edge e of $G = C_m \,\square\, C_n$ and every vertex w not on the cycle of G (column or row) through e, the coefficient of $x_w \prod_{v \neq w} x_v^2$ in P_{G-e} is nonzero. Hence if some edge uv has $L(u) \cap L(v) = \emptyset$, then G is L-colourable.*

The proof uses the 30×30 (for $m = 5$) column transfer matrix: it is diagonalisable, a unique pair $\pm i\lambda_1$ ($\lambda_1 \approx 5.6232$) dominates, the dominant amplitudes are exactly nonzero, and the finitely many small n are computed exactly. The case $m = 3$ is Amrutanshu's. So in a bad 3-list assignment, every edge has overlapping lists.

Computation 7.3 (W2)**.** There is no list assignment of $C_5 \,\square\, C_5$ with lists of size 3 from a palette of 4 colours that has no colouring. (Counterexample-guided search; the final master problem is certified unsatisfiable by an independent solver, and every learned clause and each of its 4 800 symmetry images is re-verified. The same holds for $C_3 \,\square\, C_5$, as it must by Theorem A.)

7.2 The obstacle: two-column extension for 5-cycles

The proof of Theorem A generalises step by step except one. Pick $\alpha \in L(r, j) \setminus L(r, j+1)$, delete columns $j+1, j+2$, and colour the strip $C_m \,\square\, P_{n-2}$. The strip has mk vertices and $2mk - m$ edges ($k = n - 2$), so the Nullstellensatz leaves m units of freedom at its ends: 2 to pin (r, j) to α, and 1 to forbid a colour β_t at each row t of column $j+3$ outside a pair $\{p, q\}$.

Strip coefficients. Amrutanshu observed, for $m \in \{3, 5, 7, 9\}$ and $k \leq 9$, that the coefficient of this shape vanishes exactly when k is odd and $2r \equiv p + q \pmod{m}$. The zeros are forced: the row reflection fixing r swaps p and q, fixes the shape, and reverses the mk column edges, an odd number. We confirmed the rule exactly for $m = 5$ and every strip length from 2 to 27 (2 600 cases, no exception).

The extension. What remains is to fill $C_m \,\square\, K_2$ (the two deleted columns). For $m = 3$ this is Lemma 4.1. For $m = 5$ we model a *window*: lists of size 3 on the two deleted columns, the pin at row 0, and an adversary who chooses any proper colouring of the 5-cycles in columns j and $j+3$ (with $(0, j)$ coloured α). A *shape* is an end weight with $c_S \neq 0$; it lets us forbid one colour at chosen rows. A window is *good* if some shape and some forbidden colours make the fill succeed against every adversary.

Computation 7.4 (W8, $m = 5$, palette of 4 colours)**.** (a) The plan's shape (forbid at the rows of column $j+3$ outside $\{p, q\}$, $2r \not\equiv p + q$) fails on 6 904 of the $4^{10} = 1\,048\,576$ windows. Random windows from 5 or 6 colours (60 000) never fail.

(b) There are 74 end weights with the pin at row 0, all other weights 0 or 1, and $c_S \neq 0$ for *every* odd strip length from 3 to 27; 66 of them forbid colours on the α side. Allowing all 74 leaves 400 bad windows, and 184 once α is taken to be the actual colour (with 4 colours it is forced).

(c) A cut can be made forwards (delete $j+1, j+2$) or backwards (delete $j, j-1$, using $\alpha' \in L(r, j+1) \setminus L(r, j)$), at any row r and column j with $L(r, j) \neq L(r, j+1)$. For $n \in \{5, 7\}$, every 3-list assignment of $C_5 \,\square\, C_n$ from 4 colours in which *every* cut is bad is row-constant, column-constant, or has four rows with one common constant list: there are exactly 600 (for $n = 5$) and 2 520 (for $n = 7$) assignments of the last kind, and no others. (For $n = 9$ we have only checked that such assignments exist.)

The last kind is coloured directly.

Lemma 7.5 (One free row)**.** *Let $m \geq 5$ be odd and $n \geq 3$. If L is a 3-list assignment of $C_m \,\square\, C_n$ and there are a row i_0 and a set A with $L(i,j) = A$ for all $i \neq i_0$ and all j, then $C_m \,\square\, C_n$ is L-colourable.*

Proof. The rows other than i_0 form a path $R = (i_0+1, \ldots, i_0-1)$ of $m-1 \geq 4$ rows, an even number. Choose a proper colouring $p : R \to \mathbb{Z}_3$ of this path whose two ends get the same value: $0, 1, 2, 0$ followed by $1, 0, 1, 0, \ldots$ for longer paths. Let $\rho(x) = x + 1$ on $\mathbb{Z}_3$. Choose $\sigma_0, \ldots, \sigma_{n-1} \in \{\rho, \rho^{-1}\}$ with $\sigma_0 \cdots \sigma_{n-1} = \mathrm{id}$; this needs k copies of ρ with $2k \equiv n \pmod 3$, and $k \in \{0, 1, 2\}$ works. Fix a bijection $\pi_0 : \mathbb{Z}_3 \to A$, put $\pi_{j+1} = \pi_j \sigma_j$, and colour (i, j) with $\pi_j(p(i))$ for $i \in R$. Columns are proper because p is, and row edges are proper because σ_j has no fixed point. The two neighbours of (i_0, j) in column j get the same colour $a_j := \pi_j(p(\mathrm{end}))$, so (i_0, j) keeps at least 2 colours of $L(i_0, j) \setminus \{a_j\}$, and row i_0 is a cycle with lists of size at least 2.

Such a cycle is colourable unless it is odd and all its lists are one 2-set B [ERT]. Suppose this happens. Then $L(i_0, j) = B \cup \{a_j\}$ with $a_j \notin B$ for every j, and since $a_{j+1} \neq a_j$, the colours a_j take at least two values in $A \setminus B$. Now replace every π_j by $\tau\pi_j$ for a 3-cycle τ of A. This is again a valid colouring of the rows in R, with a_j replaced by $\tau(a_j) \neq a_j$. If row i_0 fails again, then $\tau(a_j) \in L(i_0, j) \setminus \{a_j\} = B$ for all j. So τ maps two distinct colours of $A \setminus B$ into $A \cap B$, and $|A \cap B| \geq 2$, which forces $B \subseteq A$ and $|A \setminus B| = 1$, a contradiction. □

Theorem 7.6 (Odd strip lemma for $m = 5$; computer-assisted, W7)**.** *For each of the* 74 *end weights μ of Computation 7.4(b), in either orientation of the strip, the coefficient $c_\mu(\ell)$ of $C_5 \,\square\, P_\ell$ is nonzero for* every *odd $\ell \geq 3$. Hence the strip can be coloured with the pin and the forbidden colours of any of these shapes, at every odd length.*

Proof sketch. $c_\mu(\ell) = u^{\mathsf{T}} T^{\ell-2} v$ with the 30×30 transfer matrix T of Theorem 7.2, whose minimal polynomial is $x\, h_1(x^2)\, h_2(x^2)$ with $h_1 = y^2 + 10y + 5$ and $h_2 = y^4 + 35y^3 + 110y^2 + 100y + 25$, both irreducible over $\mathbb{Q}$ with only negative real roots. So $s_j = c_\mu(2j+1)$ satisfies the recurrence $h_1 h_2$ for $j \geq 1$, and its minimal recurrence over $\mathbb{Q}$ is h_1, h_2 or $h_1 h_2$ (found exactly from $s_1, \ldots, s_{13}$; this suffices since $13 - 6 \geq 6$). It is h_2 for 96 and h_1 for 52 of the 148 cases. Minimality over $\mathbb{Q}$ is minimality over $\mathbb{C}$, so every root has a nonzero amplitude. The root of largest modulus is unique, and certified ball arithmetic shows that its term outweighs all others already at $j = 1$, hence for all $j \geq 1$. The predicted recurrence was also confirmed on exact values up to $\ell = 41$. (Script `ws-W7/prove_strip5.py`.) □

This settles item A1 of the work plan for $m = 5$ (the midpoint rule's nonvanishing for all odd lengths) and more. Putting the pieces together:

Theorem 7.7 (Reduction for $m = 5$)**.** *Let $n \geq 5$ be odd and L a 3-list assignment of $C_5 \,\square\, C_n$. Then $C_5 \,\square\, C_n$ is L-colourable if L is row-constant, column-constant, has one free row, or has a good cut.*

Proof. The first two cases are Corollary 7.1(b), the third is Lemma 7.5. A good cut at a trigger (r, j) uses a shape whose strip coefficient is nonzero at length $n - 2$ (Theorem 7.6); the strip colouring it produces is one of the adversary's options, and goodness says the two deleted columns can then be filled. □

So the whole of $C_5 \,\square\, C_n$ rests on one statement.

Open Problem 7.8 (Window statement)**.** Every 3-list assignment of $C_5 \,\square\, C_n$ (n odd) in which every cut is bad is row-constant, column-constant, or has one free row.

By Computation 7.4(c) this holds for all assignments from 4 colours and $n \in \{5, 7\}$, so Theorem 7.7 proves those cases. With more colours the window picture is richer: changing a single list of a bad 4-colour window to a 3-subset of 6 colours gives 992 further bad windows (with α fresh), for example $L_1 = (\{0,1,3\}^3, \{0,3,4\}, \{0,1,3\})$, $L_2 = (\{0,1,3\}^3, \{0,2,3\}, \{0,1,3\})$. Evaluated for each admissible α separately (4 728 bad pairs of window and α), these windows still produce no new obstruction: for $n = 5$ the assignments in which every cut is bad are again only column-constant (24) or one free row (39 120, counting cyclic shifts). This covers only the windows within one changed list of a bad 4-colour window. An exhaustive check of all 6.7×10^8 window types with at most 5 colours is about 370 CPU-hours (a day on an 18-core machine); a proof for all palettes needs an argument, not enumeration.

8 Verification

All paths are relative to `odd-torus/`.

Claim	Script	Output
Strip coefficient closed form, ℓ from 2 to 11	`ws-W1/strip_coeff.py`, `direct_expand.py`, `compare.py`	0 mismatches in 500
Lemma 4.1	`ws-W9/prism_sizes.c`	8 335 non-colourable, 0 exceptions
Prism lemma (Z)/(W)	`ws-W9/prism.c`	986 non-colourable, 0 mismatches
Case 1 extension, palettes 5, 6	`ws-W9/ext3.c`	0 failures
Control (no β)	`ws-W9/ext3_nododge.c`	14 592 failures
Theorem 7.2	`ws-W4/`	see `ws-W4/REPORT.md`
Computation 7.3	`ws-W2/certify.py`	`certify_t55q4_nosym.txt`
Midpoint rule, $m = 5$	`ws-W9/midpoint.py`	0 mismatches in 2 600
Windows, plan shape, $m = 5$	`ws-W8/window5.c`	6 904 bad of 1 048 576
Windows, all 74 shapes	`ws-W8/window5g.c`, `shapes5.py`	400; 184 with real α
Window code cross-check	`ws-W8/validate_window.py`	agrees on 6 windows
Lemma 7.5 construction	`ws-W9/freerow_check.py`	0 failures in 20 000
Theorem 7.6	`ws-W7/prove_strip5.py`	148 cases proved
Bad windows, 5–6 colours	`ws-W8/nbhd.py`, `window5`, `window5g`	992 (one list changed)
Every cut bad, 5–6 colours, $n = 5$	`ws-W8/global_alpha.py`	only column-constant, one free row
Every cut bad, $n = 5, 7$	`ws-W8/global_q4_all_n.py`	600 and 2 520, all one free row
Kernel orientation counts	`ws-W5/`	see `ws-W5`

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